Therefore $a = -2/b$.
From $a^2 + 3b^2 = 7$, we get $(-2/b)^2 + 3b^2 = 7$, which gives $4/b^2 + 3b^2 = 7$ or $3b^4 - 7b^2 + 4 = 0$.
Solving for b^2 we get $b^2 = \frac{(7 \pm \sqrt{49-48})}{6}$, which gives $b^2 = 1$ or $b^2 = \frac{4}{3}$.
Thus, $b = \pm 1$ or $b = \pm\frac{2}{\sqrt{3}}$.
Since b is rational, we have $b = \pm 1$ and hence the corresponding values of a are ∓ 2.
Since $\sqrt{7 - 4\sqrt{3}} > 0$, we have $\sqrt{7 - 4\sqrt{3}} = 2 - \sqrt{3}$.



It is not always possible to express square roots of $u + v\sqrt{b}$ where u, v are rationals, in the form $x + y\sqrt{b}$ with x, y rationals. For example, the square root of $1 + \sqrt{2}$ is not of the form $a + b\sqrt{2}$ with a, b rationals.



Exercise - 2.11

1. Simplify:

   (i) $(125)^{\frac{2}{3}}$, (ii) $16^{\frac{-3}{4}}$, (iii) $(-1000)^{\frac{-2}{3}}$, (iv) $(3^{-6})^{\frac{1}{3}}$, (v) $\dfrac{27^{\frac{-2}{3}}}{27^{\frac{-1}{3}}}$.

2. Evaluate $\left(\left((256)^{-1/2}\right)^{\frac{-1}{4}}\right)^3$.
3. If $(x^{1/2} + x^{-1/2})^2 = 9/2$, then find the value of $(x^{1/2} - x^{-1/2})$ for $x > 1$.
4. Simplify and hence find the value of n: $3^{2n}9^2 3^{-n}/3^{3n} = 27$.
5. Find the radius of the spherical tank whose volume is $32\pi/3$ units.
6. Simplify by rationalising the denominator. $\dfrac{7 + \sqrt{6}}{3 - \sqrt{2}}$.
7. Simplify $\dfrac{1}{3 - \sqrt{8}} - \dfrac{1}{\sqrt{8} - \sqrt{7}} + \dfrac{1}{\sqrt{7} - \sqrt{6}} - \dfrac{1}{\sqrt{6} - \sqrt{5}} + \dfrac{1}{\sqrt{5} - 2}$.
8. If $x = \sqrt{2} + \sqrt{3}$ find $\dfrac{x^2 + 1}{x^2 - 2}$.

2.9 Logarithm

We have seen that, with a base $0 < a \neq 1$, the exponential function $f(x) = a^x$ is defined on $\mathbb{R}$ having range $(0, \infty)$. We also observed that $f(x)$ is a bijection, hence it has an inverse. We call this inverse function as ***logarithmic function*** and is denoted by $\log_a(.)$. Let us discuss this function further. Note that if $f(x)$ takes x to $y = a^x$, then $\log_a(.)$ takes y to x. That is, for $0 < a \neq 1$, we have

$$y = a^x \qquad \text{is equivalent to} \qquad \log_a y = x.$$

For example, since $3^4 = 81$ we have $\log_3(81) = 4$. In other words, with fixed a, given a real number y, logarithm finds the exponent x satisfying $a^x = y$. This is useful in addressing practical problems like, "how long will it take for certain investment to reach a fixed amount?" Logarithm is also very useful in multiplying very small or big numbers.



(i) Note that exponential function a^x is defined for all $x \in \mathbb{R}$ and $a^x > 0$ and so $\log_a(\cdot)$ defined **only for positive real numbers**.
(ii) Also, $a^0 = 1$ for any base a and hence $\log_a(1) = 0$ for any base a.